

Topic 15: Linear Systems of ODEs & Phase Plane Portraits

Professor Leonard Master Series · Linear Systems

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Coupled Dynamics

In matrix form:

$$\mathbf{X}'(t) = \mathbf{A}\mathbf{X}(t)$$

We seek straight-line trajectories along which the velocity vector $\mathbf{X}'$ points directly parallel to the position vector $\mathbf{X}$:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \iff (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

2. Curriculum Reference Diagram

corresponding entries in the vector of the left, we obtain the more familiar statement

$$x = c_1e^{-t} + 3c_2e^{4t}, y = -c_1e^{-t} + 2c_2e^{4t}.$$

As pointed out in Section 10.1, we can interpret these equations as parametric equations of a curve or **trajectory** in the xy -plane or **phase plane**. The three graphs shown in **FIGURE 10.2.1**, $x(t)$ in the tx -plane, $y(t)$ in the ty -plane, and the trajectory in the phase plane, correspond to the choice of constants $c_1 = c_2 = 1$ in the solution. A collection of trajectories in the phase plane as shown in **FIGURE 10.2.2** is said to be a **phase portrait** of the given linear system. What appears to be two red lines in Figure 10.2.2 are

Figure 15.1: Trajectories in phase plane for 2x2 linear system — from Zill 7th Ed. Chapter 10 (Fig. 10.2.2).

3. Phase Plane Zoo: The Geometrical Classification

Compute λ_1, λ_2 from $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$:

- **Both Real, Negative ($\lambda_1, \lambda_2 < 0$): Stable Node (Sink)** — All trajectories rush into the origin.
- **Both Real, Positive ($\lambda_1, \lambda_2 > 0$): Unstable Node (Source)** — All trajectories flee away.
- **Opposite Signs ($\lambda_1 < 0 < \lambda_2$): Saddle Point (Always Unstable)** — Trajectories approach along the stable eigenvector and get flung away along the unstable eigenvector!
- **Pure Imaginary ($\lambda = \pm i\beta$): Center (Neutrally Stable)** — Closed concentric ellipses.
- **Complex with Real Part ($\lambda = \alpha \pm i\beta$):**
 - $\alpha < 0$: **Stable Spiral Sink** (spirals inward).
 - $\alpha > 0$: **Unstable Spiral Source** (spirals outward).